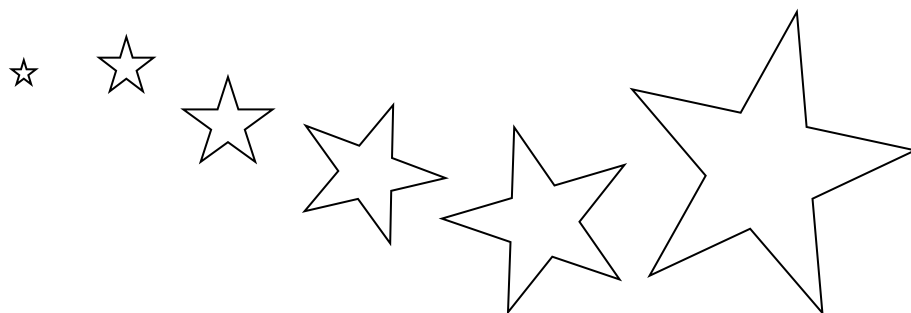




3 Unit Selective
Questions

Induction

- Ⓢ = Summation
- ⓓ = Divisibility
- Ⓜ = Miscellaneous



Name: _____

Solutions



Question 1



WESTERN

Use the process of mathematical induction to show that:

$$1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1)$$

b. $1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1)$

Test $n = 1$. $3^{1-1} = \frac{1}{2}(3^1 - 1) = 1 \quad \therefore \text{true for } n = 1$

$n = k$ i.e. $1 + 3 + 9 + \dots + 3^{k-1} = \frac{1}{2}(3^k - 1)$

Prove true for $n = k + 1$

i.e.

$$\begin{aligned} 1 + 3 + 9 + \dots + 3^{k-1} + 3^k &= \frac{1}{2}(3^k - 1) + 3^{(k+1)-1} \\ &= \frac{1}{2}(3^k - 1) + 3^k \\ &= \frac{1}{2}(3^k - 1 + 2 \cdot 3^k) \\ &= \frac{1}{2}(3 \cdot 3^k - 1) \\ &= \frac{1}{2}(3^{k+1} - 1) \end{aligned}$$

Which is in the form $\frac{1}{2}(3^n - 1)$ where $n = k + 1$

$\therefore$ True for $n = k + 1$ when true for $n = k$,

But it is true for $n = 1$

$\therefore$ True for $n = 1 + 1 = 2$

And $2 + 1 = 3$

Etc.

Hence by Mathematical Induction

$$1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1)$$



Question 2



CSSA

Use Mathematical Induction to show that $5^n + 2(11^n)$ is a multiple of 3 for all positive integers n .

5

Define the sequence of statements $S(n)$: $5^n + 2(11^n)$ is a multiple of 3, $n = 1, 2, 3, \dots$

Consider $S(1)$: $5^1 + 2(11^1) = 27 = 3 \times 9 \quad \therefore S(1)$ is true.

If $S(k)$ is true, then $5^k + 2(11^k) = 3M$ for some integer M . **

Consider $S(k+1)$: $5^{k+1} + 2(11^{k+1}) = 5(5^k) + 22(11^k) = 5\{5^k + 2(11^k)\} + 12(11^k)$

$\therefore 5^{k+1} + 2(11^{k+1}) = 5(3M) + 12(11^k) = 3\{5M + 4(11^k)\}$ if $S(k)$ is true, using **

But M and k integral $\Rightarrow \{5M + 4(11^k)\}$ is an integer.

$\therefore S(k)$ true $\Rightarrow S(k+1)$ true, $k = 1, 2, 3, \dots$

Hence $S(1)$ is true, and if $S(k)$ is true, then $S(k+1)$ is true. $\therefore S(2)$ is true, and then $S(3)$ is true, and so on. Hence by Mathematical Induction, $S(n)$ is true for all positive integers n .



Question 3



ABBOTSLEIGH

Use mathematical induction to prove that for all integers $n = 1, 2, 3, \dots$

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$$

b) Show that the result is true for $n=1$

$$LHS = 1 \times 1!$$

$$= 1$$

$$RHS = (1+1)! - 1$$

$$= 2! - 1$$

$$= 1$$

$$= LHS \quad \therefore \text{result is true for } n=1$$

Assume that the result is true for $n=k$

i.e. assume $1 \times 1! + 2 \times 2! + \dots + k \times k! = (k+1)! - 1$

Using this assumption, show that the result is true for $n=k+1$.

i.e. show $1 \times 1! + 2 \times 2! + \dots + k \times k! + (k+1)(k+1)!$

$$= (k+2)! - 1 \quad \rightarrow$$

is true

$$LHS = (k+1)! - 1 + (k+1)(k+1)!$$

$$= (k+1)! (1 + k+1) - 1$$

$$= (k+1)! (k+2) - 1$$

$$= (k+2)! - 1$$

$$= RHS$$

$\therefore$ if the result is true for $n=k$ then it is also true for $n=k+1$

By the process of mathematical induction, this result is true for all integers $n = 1, 2, 3, \dots$



Question 4



INDEPENDENT

Use the method of mathematical induction to prove that $(1+1) + (2+3) + (3+5) + \dots + (n + (2n-1)) = \frac{1}{2}n(3n+1)$ where n is a positive integer.

$$(1+1) + (2+3) + (3+5) + \dots + (n + (2n-1)) = \frac{1}{2}n(3n+1), \text{ } n \text{ is a positive integer}$$

Step 1

For $n = 1$,

$$\begin{aligned} LHS &= 1+1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} RHS &= \frac{1}{2}(1)(3+1) \\ &= 2 \end{aligned}$$

$$\therefore LHS = RHS$$

Hence true for $n = 1$

Step 2: Assume true for $n = k$,

$$(1+1) + (2+3) + (3+5) + \dots + (k + (2k-1)) = \frac{1}{2}k(3k+1) \quad (*)$$

Prove true for $n = k+1$,

$$\begin{aligned} LHS &= (1+1) + (2+3) + \dots + (k + (2k-1)) + (k+1 + (2(k+1)-1)) \\ &= \frac{1}{2}k(3k+1) + (k+1 + 2k+1) \quad (\text{from } *) \\ &= \frac{3}{2}k^2 + \frac{1}{2}k + 3k + 2 \\ &= \frac{3}{2}k^2 + \frac{7}{2}k + 2 \\ &= \frac{1}{2}(3k^2 + 7k + 4) \\ &= \frac{1}{2}(k+1)(3k+4) \\ &= \frac{1}{2}(k+1)(3(k+1)+1) \end{aligned}$$

So it is true for $n = (k+1)$

Step 3 Since it is true for $n = 1, n = 2$ and so on and it is true for $n = k+1$ by assuming true for $n = k$, therefore by mathematical induction, it is true for all n where n is a positive integer.



Question 5



INDEPENDENT

Use Mathematical Induction to show that, for all positive integers $n \geq 1$,
 $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$.

Let $S(n)$ be the sequence of statements $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$, $n = 1, 2, 3, \dots$

Consider $S(1)$: $LHS = 1^2 = 1$ and $RHS = \frac{1}{3} \cdot 1 \cdot (2-1)(2+1) = 1$. Hence $S(1)$ is true

If $S(k)$ is true : $1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1)$ **

Consider $S(k+1)$: $LHS = \left\{ 1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 \right\} + (2k+1)^2$
 $= \frac{1}{3}k(2k-1)(2k+1) + (2k+1)^2$ if $S(k)$ is true, using **
 $= \frac{1}{3}(2k+1)\{k(2k-1) + 3(2k+1)\}$
 $= \frac{1}{3}(2k+1)\{2k^2 + 5k + 3\}$
 $= \frac{1}{3}(2k+1)(k+1)(2k+3)$
 $= \frac{1}{3}(k+1)\{2(k+1)-1\}\{2(k+1)+1\}$
 $= RHS$

Hence if $S(k)$ is true, then $S(k+1)$ is true. But $S(1)$ is true, hence $S(2)$ is true, and then $S(3)$ is true and so on. Therefore $S(n)$ is true for all positive integers $n \geq 1$.



Question 6



CSSA

Use Mathematical Induction to show that for all positive integers $n \geq 1$,

4

$$\frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+n} = \frac{2n}{n+1}.$$

Let $S(n)$ be the statement $\frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+n} = \frac{2n}{n+1}$, $n=1, 2, 3, \dots$

Consider $S(1)$: $LHS = \frac{1}{1} = 1$; $RHS = \frac{2 \times 1}{1+1} = 1$. Hence $S(1)$ is true.

If $S(k)$ is true: $\frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+k} = \frac{2k}{k+1}$ **

Consider $S(k+1)$:

$$\begin{aligned} LHS &= \frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+k} + \frac{1}{1+2+3+\dots+(k+1)} \\ &= \frac{2k}{k+1} + \frac{1}{1+2+3+\dots+(k+1)} \quad \text{if } S(k) \text{ is true, using **} \\ &= \frac{2k}{k+1} + \frac{2}{(k+1)\{1+(k+1)\}} \quad \text{sum of } (k+1) \text{ terms of A.P: } a=1, d=1 \\ &= \frac{2}{(k+1)} \left\{ k + \frac{1}{k+2} \right\} \\ &= \frac{2(k^2 + 2k + 1)}{(k+1)(k+2)} \\ &= \frac{2(k+1)^2}{(k+1)(k+2)} \\ &= \frac{2(k+1)}{\{(k+1)+1\}} \\ &= RHS \end{aligned}$$

Hence if $S(k)$ is true, then $S(k+1)$ is true. But $S(1)$ is true, hence $S(2)$ is true, and then $S(3)$ is true and so on. Therefore by Mathematical Induction, $S(n)$ is true for all positive integers n .



Question 7



CSSA

Use Mathematical Induction to show that for all positive integers $n \geq 1$

4

$$\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \dots + \frac{n+2}{n(n+1)2^n} = 1 - \frac{1}{(n+1)2^n}.$$

Let $S(n)$, $n=1, 2, 3, \dots$ be the sequence of statements $\sum_{r=1}^n \frac{r+2}{r(r+1)2^r} = 1 - \frac{1}{(n+1)2^n}$, $n=1, 2, 3, \dots$

Consider $S(1)$: $LHS = \frac{3}{1 \times 2 \times 2} = \frac{4-1}{2 \times 2^1} = 1 - \frac{1}{2 \times 2^1} = RHS$. $\therefore S(1)$ is true.

If $S(k)$ is true: $\sum_{r=1}^k \frac{r+2}{r(r+1)2^r} = 1 - \frac{1}{(k+1)2^k}$ **

Consider $S(k+1)$: $LHS = \sum_{r=1}^{k+1} \frac{r+2}{r(r+1)2^r}$

$$= \sum_{r=1}^k \frac{r+2}{r(r+1)2^r} + \frac{(k+1)+2}{(k+1)(k+2)2^{k+1}}$$

$$= 1 - \frac{1}{(k+1)2^k} + \frac{(k+1)+2}{(k+1)(k+2)2^{k+1}} \quad \text{if } S(k) \text{ is true, using **}$$

$$= 1 - \frac{2(k+2) - (k+3)}{(k+1)(k+2)2^{k+1}}$$

$$= 1 - \frac{k+1}{(k+1)(k+2)2^{k+1}}$$

$$= 1 - \frac{1}{(k+2)2^{k+1}}$$

$$= RHS$$

Hence if $S(k)$ is true, then $S(k+1)$ is true. But $S(1)$ is true, hence $S(2)$ is true, and then $S(3)$ is true and so on. Hence by mathematical induction $S(n)$ is true for all positive integers $n \geq 1$.



Question 8



CSSA

Prove by mathematical induction that $\sum_{r=1}^n r \times r! = (n+1)! - 1$.

Let $S(n)$ be the statement $\sum_{r=1}^n r \times r! = (n+1)! - 1$

Consider $S(1)$: $LHS = 1 \times 1!$; $RHS = (1+1)! - 1 = 1$.

Hence $S(1)$ is true

If $S(k)$ is true: $\sum_{r=1}^k r \times r! = (k+1)! - 1$ *

RTP $S(k+1)$ is true i.e. to prove $\sum_{r=1}^{k+1} r \times r! = (k+2)! - 1$

$$\begin{aligned} LHS &= \sum_{r=1}^k r \times r! + (k+1)(k+1)! \\ &= (k+1)! - 1 + (k+1)(k+1)! \quad \text{if } S(k), \text{ using } * \\ &= (k+1)! (1 + k + 1) - 1 \\ &= (k+1)! (k+2) - 1 \end{aligned}$$

Hence if $S(k)$ then $S(k+1)$ is true. Thus since $S(1)$ is true it follows by induction that $S(n)$ is true for positive integral n .



Question 9



INDEPENDENT

Use Mathematical Induction to show that for all positive integers n ,

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n.$$

Define the sequence of statements $S(n)$, $n = 1, 2, 3, \dots$ by

$$S(n): 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n$$

Consider $S(1)$: $LHS = 1 \times 2^0 = 1$, $RHS = 1 + 0 \times 2^1 = 1$ $\therefore S(1)$ is true

If $S(k)$ is true: $1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1)2^k$ **

Consider $S(k+1)$: $LHS = (1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1}) + (k+1) \times 2^k$
 $= (1 + (k-1)2^k) + (k+1)2^k$ if $S(k)$ is true, using **
 $= 1 + 2k \cdot 2^k$
 $= 1 + k \cdot 2^{k+1}$
 $= 1 + ((k+1) - 1)2^{k+1}$
 $= RHS$

Hence if $S(k)$ is true, then $S(k+1)$ is true. But $S(1)$ is true, hence $S(2)$ is true, and then $S(3)$ is true and so on. $\therefore S(n)$ is true for all positive integers n .



Question 10



Asquith Boys High School

Prove by the method of Mathematical Induction that

4

$$\sum_{r=1}^n 5^{r-1} = \frac{5^n - 1}{4}$$

$$\sum_{r=1}^n 5^{r-1} = \frac{5^n - 1}{4}$$

Let $n=1$

$$\text{LHS} = 5^{0-1} = 1$$

$$\text{RHS} = \frac{5^1 - 1}{4} = 1$$

Assume true for $n=k$.

$$5^{0-1} + 5^{1-1} + 5^{2-1} + \dots + 5^{k-1} = \frac{5^k - 1}{4}$$

Prove for $n=k+1$

$$\text{HS} = \frac{5^k - 1}{4} + 5^{k-1}$$

$$\text{RHS} = \frac{5^{k+1} - 1}{4}$$

$$= \frac{5^k - 1 + 4 \cdot 5^{k-1}}{4}$$

$$= \frac{5^k [1+4] - 1}{4}$$

$$= \frac{5^{k+1} - 1}{4}$$

$$= \text{RHS}$$

If true for $n=k$, and $n=k+1$, then true for $n \in \mathbb{N}$ integers by the principles of Mathematical Induction.



Question 11



ARC

Use mathematical induction to show that for all positive

integers n , $\sum_{r=1}^n a^{-r} = \frac{a^n - 1}{(a-1)a^n}$

(4 marks)

$$\sum_{r=1}^n a^{-r} = \frac{1}{a} + \frac{1}{a^2} + \dots + \frac{1}{a^n}$$

We are required to prove that

$$\frac{1}{a} + \frac{1}{a^2} + \dots + \frac{1}{a^n} = \frac{a^n - 1}{(a-1)a^n} \text{ for all positive integers } n$$

Test $n = 1$:

$$LHS = \frac{1}{a}$$

$$\begin{aligned} RHS &= \frac{a-1}{(a-1)a} \\ &= \frac{1}{a} \end{aligned}$$

$\therefore$ The result is true for $n = 1$

Assume that it is true for $n = k$

$$\frac{1}{a} + \frac{1}{a^2} + \dots + \frac{1}{a^k} = \frac{a^k - 1}{(a-1)a^k}$$

Put $n = k + 1$

$$\begin{aligned} LHS &= \frac{1}{a} + \frac{1}{a^2} + \dots + \frac{1}{a^k} + \frac{1}{a^{k+1}} \\ &= \frac{a^k - 1}{(a-1)a^k} + \frac{1}{a^{k+1}} \\ &= \frac{a(a^k - 1) + (a-1)}{(a-1)a^{k+1}} \\ &= \frac{a^{k+1} - a + a - 1}{(a-1)a^{k+1}} \\ &= \frac{a^{k+1} - 1}{(a-1)a^{k+1}} \end{aligned}$$

Hence if the result is true for $n = k$, it is true for $n = k + 1$.

But it is true for $n = 1$. Therefore it is true for $n = 2$ and so, by mathematical induction, for all positive integers n .



Question 12



THG

Use the method of mathematical induction to prove that $(1+1) + (2+3) + (3+5) + \dots + (n + (2n-1)) = \frac{1}{2}n(3n+1)$ where n is a positive integer.

$$(1+1) + (2+3) + (3+5) + \dots + (n + (2n-1)) = \frac{1}{2}n(3n+1), \text{ } n \text{ is a positive integer}$$

Step 1

For $n = 1$,

$$\begin{aligned} LHS &= 1+1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} RHS &= \frac{1}{2}(1)(3+1) \\ &= 2 \end{aligned}$$

$$\therefore LHS = RHS$$

Hence true for $n = 1$

Step 2: Assume true for $n = k$,

$$(1+1) + (2+3) + (3+5) + \dots + (k + (2k-1)) = \frac{1}{2}k(3k+1) \quad (*)$$

Prove true for $n = k+1$,

$$\begin{aligned} LHS &= (1+1) + (2+3) + \dots + (k + (2k-1)) + (k+1 + (2(k+1)-1)) \\ &= \frac{1}{2}k(3k+1) + (k+1 + 2k+1) \quad (\text{from } *) \\ &= \frac{3}{2}k^2 + \frac{1}{2}k + 3k + 2 \\ &= \frac{3}{2}k^2 + \frac{7}{2}k + 2 \\ &= \frac{1}{2}(3k^2 + 7k + 4) \\ &= \frac{1}{2}(k+1)(3k+4) \\ &= \frac{1}{2}(k+1)(3(k+1)+1) \end{aligned}$$

So it is true for $n = (k+1)$

Step 3 Since it is true for $n = 1, n = 2$ and so on and it is true for $n = k+1$ by assuming true for $n = k$, therefore by mathematical induction, it is true for all n where n is a positive integer.



Question 13



CEM

Show by mathematical induction that if n is a positive integer

4

then $7^n + 2$ is divisible by 3.

Step 1: Prove true for $n = 1$

i.e. $7^1 + 2 = 9$ which is divisible by 3

Step 2 : Assume true for $n = k$

i.e. $7^k + 2 = 3M$ where M is an integer

To prove true for $n = k + 1$

i.e. $7^{k+1} + 2 = 3N$ where N is another integer

Proof : L.H.S. = $7^k \cdot 7 + 2$

$$= 7(7^k + 2) - 14 + 2$$

$$= 7(3M) - 12$$

$$= 3(7M - 4) = 3N \quad 2$$

Step 3: If it is true for $n = 1$, and also true for $n = 1 + 1 = 2$, then it is true for all positive integer value n .



Question 14



James Ruse

Prove by induction that $9^{n+2} - 4^n$ is divisible by 5 for integers $n \geq 1$.

$$9^{n+2} - 4^n \div 5.$$

Let $n=1$

$$9^3 - 4 = 725, \text{ which is divisible by } 5$$

∴ True for $n=1$ ✓

Assume true for $n=k$

$$9^{k+2} - 4^k = 5M \quad (\text{where } M \text{ is an integer}).$$

$$9^{k+2} = 5M + 4^k$$

$$9^{k+2} - 5M = 4^k$$

Prove for $n=k+1$

$$9^{k+3} - 4^{k+1}$$

$$= 9^{k+2} \cdot 9 - 4^k \cdot 4$$

$$= (5M + 4^k) \cdot 9 - 4^{k+1}$$

$$= 45M + 9 \times 4^k - 4^{k+1}$$

$$= 45M + 4^k[9 - 4]$$

$$= 45M + 4^k \cdot 5$$

$$= 5[9M + 4^k] = 5N \quad (\text{where } N \text{ is integer}).$$

which is divisible by 5.

∴ If true for $n=k$ and $n=k+1$, then by the principles of Induction
it is true for all positive integers.

Selective Questions

Induction



Question 15



James Ruse

Prove by induction that $\cos(x + n\pi) = (-1)^n \cos x$ for integer $n \geq 1$.

$$\cos(x + n\pi) = (-1)^n \cos x \quad \text{for } n \geq 1.$$

prove true for $n=1$

$$\therefore \cos(\pi + x) = (-1) \cos x.$$

$$\therefore \cos \pi \cos x - \sin \pi \sin x = -1 \cos x$$

$$-1 \cos x = -1 \cos x$$

$\therefore$ true for $n=1$.

assume true for $n=k$

$$\therefore \cos(x + k\pi) = (-1)^k \cos x.$$

prove true for $n=k+1$

$$\therefore \cos(x + (k+1)\pi) = (-1)^{k+1} \cos x.$$

take LHS.

$$\cos x \cos(k\pi + \pi) + \sin x \sin(k\pi + \pi).$$

$$= \cos x (\cos k\pi \cos \pi - \sin k\pi \sin \pi) + \sin x (\sin k\pi \cos \pi + \cos k\pi \sin \pi)$$

$$= \cos x ((-1)^k \cdot -1)$$

$$= \cos x (-1)^{k+1}$$

$$= (-1)^{k+1} \cos x$$

$$= \text{RHS}$$

$\therefore$ Since true for $n=1$ it must be true for $n=2$. Since true for $n=2$ it must be true for $n=3$ and so on. Therefore true for all $n \geq 1$.



Question 16



James Ruse

Prove by mathematical induction that $3^{4n}-1$ is divisible by 80 for all positive integers n .

d) $P(n): 3^{4n} - 1 = 80m$ (m is an integer)

Consider $P(1)$:

$$3^4 - 1 = 80 = 80(1)$$

$\therefore P(1)$ is true

Assume $P(k)$ is true;

$$3^{4k} - 1 = 80R.M. \quad \text{--- (A) } (N=k)$$

Consider $P(k+1)$:

$$3^{4k+4} - 1 = 80N \quad (N = \text{integer})$$

By (A); $3^4(3^{4k} - 1) + 3^4 - 1$

$$= 3^4(80m) + 3^4 - 1$$

$$= 6480m + 80$$

$$= 80(81m + 1)$$

$$= 80N \quad (81m + 1 \text{ is an integer})$$



Question 17



CSSA

Use mathematical induction to prove that

$$1 + 6 + 15 + \dots + n(2n-1) = \frac{1}{6}n(4n-1)(n+1)$$

for all positive integers n .

Prove that $1 + 6 + 15 + \dots + n(2n-1) = \frac{1}{6}n(4n-1)(n+1)$ using mathematical induction

$$\begin{aligned} \text{When } n=1, \quad RHS &= \frac{1}{6} \times 3 \times 2 \\ &= 1 \\ &= LHS \end{aligned}$$

So the statement is true for $n=1$.

Suppose that it is true for $n=k$.

$$\text{That is, suppose that } 1 + 6 + 15 + \dots + k(2k-1) = \frac{1}{6}k(4k-1)(k+1) \quad - (A)$$

Then, when $n=k+1$, we have

$$\begin{aligned} 1 + 6 + 15 + \dots + (k+1)\{2(k+1)-1\} &= \frac{1}{6}(k+1)\{4(k+1)-1\}\{(k+1)+1\} \\ &= \frac{1}{6}(k+1)(4k+3)(k+2) \end{aligned}$$

$$\begin{aligned} LHS &= 1 + 6 + 15 + \dots + (k+1)\{2(k+1)-1\} \\ &= 1 + 6 + 15 + \dots + k(2k-1) + (k+1)\{2(k+1)-1\} \\ &= \frac{1}{6}k(4k-1)(k+1) + (k+1)(2k+1) \quad \text{from (A)} \\ &= (k+1)\left\{\frac{1}{6}k(4k-1) + (2k+1)\right\} \\ &= (k+1)\left\{\frac{1}{6}k(4k-1) + \frac{1}{6}(12k+6)\right\} \\ &= \frac{1}{6}(k+1)\{4k^2 - k + 12k + 6\} \\ &= \frac{1}{6}(k+1)(4k+3)(k+2) \\ &= RHS \end{aligned}$$

It is true for $n=1$ and when it is true for $n=k$ it has been proven that it is true for $n=k+1$. Hence by mathematical induction it must be true for all positive integers n .



Question 18



Independent

Prove by Mathematical Induction that $3^{2n} - 1$ is divisible by 8 when n is an integer greater than 0. 4

$$S(n): 3^{2n} - 1 = 8I, \text{ where } I \text{ is an integer}$$

$$\text{First, } S(1): 3^2 - 1 = 8 = 8 \times 1$$

$\therefore S(1)$ is true

$$\text{Assume } S(k): 3^{2k} - 1 = 8I, \text{ where } I \text{ is an integer}$$

Prove $S(k+1)$.

$$\begin{aligned} \text{LHS} &= 3^{2(k+1)} - 1 \\ &= 3^{2k+2} - 1 \\ &= 3^{2k} \cdot 3^2 - 1 \\ &= (3^{2k} - 1) \cdot 9 + 8 \\ &= 8I \cdot 9 + 8 \\ &= 8(9I + 1) \end{aligned}$$

But I is an integer; so $9I + 1$ is an integer

$\therefore$ If $S(k)$ is true, $S(k+1)$ is true.

But $S(1)$ is true so $S(2)$ is true
and so on for all integers, n ,
greater than 0.



Question 19



Independent

Prove by Mathematical Induction that $5^{2n} - 1$ is divisible by 6 when n is a positive integer 4

Let $S(n): 5^{2n} - 1 = 6I$, where I is an integer.

$$S(1): \text{LHS} = 5^2 - 1 = 24 = 6 \times 4$$

$\therefore S(1)$ is true

Assume $S(k): 5^{2k} - 1 = 6I$ (I , integer)

Consider $S(k+1)$:

$$\begin{aligned} \text{LHS} &= 5^{2k+2} - 1 \\ &= 5^{2k} \cdot 5^2 - 1 \\ &= 25(5^{2k} - 1) - 1 + 25 \\ &= 25 \cdot 6I + 24 \text{ by } S(k) \\ &= 6[25I + 4] \end{aligned}$$

Now I is integer, $\therefore 25I + 4$ is integer.

Hence, if $S(k)$ is true, $S(k+1)$ is true

But $S(1)$ is true, so $S(2)$ is true, and then $S(3)$ is true and so on for all integer values of n .



Question 20



Pymble Ladies' College

Prove by Mathematical Induction that

4

$3^{2n+2} - 8n - 9$ is divisible by 64 for all positive integer n

Prove $3^{2n+2} - 8n - 9$ is divisible by 64

Step 1 When $n = 1$

$$3^{2+2} - 8 - 9 = 81 - 17 = 64$$

which is divisible by 64. (1)

Step 2. Assume that when $n = k$, $3^{2k+2} - 8k - 9$ is divisible by 64.

ie. $3^{2k+2} - 8k - 9 = 64M$ where M is an integer.

Now

$$\begin{aligned} & 3^{2(k+1)+2} - 8(k+1) - 9 \\ &= 3^{2k+4} - 8k - 8 - 9 \\ &= 3^2(3^{2k+2}) - 8k - 17 \\ &= 3^2(64M + 8k + 9) - 8k - 17 \\ &= 3^2(64M) + 64k + 64 \\ &= 64(9M + k + 1) \text{ which is divisible by 64.} \end{aligned}$$

(2)

Step 3. If it is true for $n = k$ it is proved true for $n = k+1$ and since s/m is true when $n = 1$

$\therefore$ it is also true for $n = 2, n = 3, n = 4, \dots$ & so on
Hence it is true for all positive integer of n . (1)



Question 21



NEAP

Prove, using mathematical induction, that $7^n - 4^n$ is divisible by 3, where n is a positive integer.

Prove $7^n - 4^n$ is divisible by 3.

When $n = 1$, $7^n - 4^n = 7 - 4 = 3$ which is divisible by 3

$\therefore$ it is true for $n = 1$. ✓

Assume $7^n - 4^n$ is divisible by 3 when $n = k$.

i.e. assume $7^k - 4^k = 3m$ where m is an integer. ✓

When $n = k + 1$, $7^n - 4^n = 7^{k+1} - 4^{k+1}$

$$= 7 \times 7^k - 4 \times 4^k$$

$$= 7(3m + 4^k) - 4 \times 4^k \text{ from assumption}$$

$$= 21m + 3 \times 4^k$$

$$= 3(7m + 4^k)$$

$$= 3I \text{ where } I \text{ is an integer. } \checkmark$$

$\therefore$ if $7^n - 4^n$ is divisible by 3 when $n = k$, then $7^n - 4^n$ is divisible by 3 when $n = k + 1$.

Since $7^n - 4^n$ is divisible by 3 when $n = 1$, then $7^n - 4^n$ is divisible by 3 when $n = 2, 3, \dots$

i.e. $7^n - 4^n$ is divisible by 3 for all positive integers n . ✓



Question 22



INDEPENDENT

Use the Principle of Mathematical Induction to prove that, for all positive integers, n ,

$$\sum_{r=1}^n \frac{1}{(4r-3)(4r+1)} = \frac{n}{4n+1}$$

$$(c) S(n): \sum_{r=1}^n \frac{1}{(4r-3)(4r+1)} = \frac{n}{4n+1}$$

$$S(1): LHS = \frac{1}{(4-3)(4+1)} = \frac{1}{5}$$

$$RHS = \frac{1}{4+1} = \frac{1}{5} = LHS$$

$\therefore S(1)$ is true

Assume $n=k$:

$$i.e. S(k): \sum_{r=1}^k \frac{1}{(4r-3)(4r+1)} = \frac{k}{4k+1}$$

Prove $n=k+1$

$$i.e. S(k+1): \sum_{r=1}^{k+1} \frac{1}{(4r-3)(4r+1)} = \frac{k+1}{4k+5}$$

$$LHS = \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)}$$

$$= \frac{k(4k+5) + 1}{(4k+1)(4k+5)}$$

$$= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)}$$

$$= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)}$$

$$= \frac{k+1}{4k+5} = RHS$$

If $S(k)$ is true, then $S(k+1)$ is true
 But $S(1)$ is true, so $S(2)$ is true,
 whence $S(3)$ is true and so on
 for all positive integer values
 of n .



Question 23



CSSA

Use Mathematical Induction to show that $5^n + 2(11^n)$ is a multiple of 3 for all positive integers n .

Define the sequence of statements $S(n)$: $5^n + 2(11^n)$ is a multiple of 3, $n = 1, 2, 3, \dots$

Consider $S(1)$: $5^1 + 2(11^1) = 27 = 3 \times 9 \quad \therefore S(1)$ is true.

If $S(k)$ is true, then $5^k + 2(11^k) = 3M$ for some integer M . **

Consider $S(k+1)$: $5^{k+1} + 2(11^{k+1}) = 5(5^k) + 22(11^k) = 5\{5^k + 2(11^k)\} + 12(11^k)$

$\therefore 5^{k+1} + 2(11^{k+1}) = 5(3M) + 12(11^k) = 3\{5M + 4(11^k)\}$ if $S(k)$ is true, using **

But M and k integral $\Rightarrow \{5M + 4(11^k)\}$ is an integer.

$\therefore S(k)$ true $\Rightarrow S(k+1)$ true, $k = 1, 2, 3, \dots$

Hence $S(1)$ is true, and if $S(k)$ is true, then $S(k+1)$ is true. $\therefore S(2)$ is true, and then $S(3)$ is true, and so on. Hence by Mathematical Induction, $S(n)$ is true for all positive integers n .



Question 24



NEAP

Use mathematical induction to prove that

$$\log 2 + \log\left(\frac{3}{2}\right) + \log\left(\frac{4}{3}\right) + \dots + \log\left(\frac{n+1}{n}\right) = \log(n+1) \text{ for } n \geq 1.$$

(a) To be proved:

$$\log 2 + \log\left(\frac{3}{2}\right) + \log\left(\frac{4}{3}\right) + \dots + \log\left(\frac{n+1}{n}\right)$$

$$= \log(n+1), n \geq 1$$

Step 1

Let $n = 1$.

The statement becomes $\log 2 = \log 2$ which is true.

Step 2

Assume the statement is true for some positive integer k .

That is, assume

$$\log 2 + \log\left(\frac{3}{2}\right) + \log\left(\frac{4}{3}\right) + \dots + \log\left(\frac{k+1}{k}\right)$$

$$= \log(k+1) \quad [\text{statement 1}]$$

Now it has to be shown that the following statement, where $n = k+1$, is true.

$$\log 2 + \log\left(\frac{3}{2}\right) + \log\left(\frac{4}{3}\right) + \dots + \log\left(\frac{k+1}{k}\right) + \log\left(\frac{k+2}{k+1}\right)$$

$$= \log(k+2) \quad [\text{statement 2}]$$

$$\text{Now } \log 2 + \log\left(\frac{3}{2}\right) + \log\left(\frac{4}{3}\right) + \dots + \log\left(\frac{k+1}{k}\right) + \log\left(\frac{k+2}{k+1}\right)$$

$$= \log(k+1) + \log\left(\frac{k+2}{k+1}\right) \quad (\text{by statement 1 above})$$

$$= \log(k+1) + \log(k+2) - \log(k+1) \quad (\text{by logarithm laws})$$

$$= \log(k+2), \text{ as required.}$$

Hence if statement 1 is true, then so is statement 2.

Now, if the original statement is true when $n = k$, where k is a positive integer, then it is also true when $n = k+1$. But

the statement is true when $n = 1$. Therefore, the statement is true for all positive integers n .



Question 25



SG

Use mathematical induction to prove that, for all integers n with $n \geq 1$

$$3 \cdot 2! + 7 \cdot 3! + 13 \cdot 4! + \dots + (n^2 + n + 1)(n + 1)! = n(n + 2)!$$

(a) Step 1: For $n=1$, $3 \cdot 2! = 1(1+2)!$

$\therefore 6=6$ Hence statement is true

for $n=1$.

Step 2: Assume that the statement is true for $n=k$

$$3 \cdot 2! + 7 \cdot 3! + \dots + (k^2 + k + 1)(k + 1)!$$

$$= k(k + 2)! \quad \text{①}$$

Our aim is to prove it true for $n=k+1$

$$\text{i.e. } 3 \cdot 2! + 7 \cdot 3! + \dots + [(k+1)^2 + k + 2](k + 2)!$$

$$= (k+1)(k+3)!$$

Starting from ① and adding

$[(k+1)^2 + k + 2](k + 2)!$ to both sides:

$$3 \cdot 2! + 7 \cdot 3! + \dots + [(k+1)^2 + k + 2](k + 2)!$$

$$= k(k + 2)! + [(k+1)^2 + k + 2](k + 2)!$$

$$\therefore \text{LHS} = [k^2 + 4k + 3](k + 2)! \quad (\text{factorizing})$$

$$= (k+1)(k+3)(k + 2)!$$

$$= (k+1)(k+3)!(\text{Since } (k+3)(k + 2)! \times (k + 2))$$

Hence if the statement is true for $n=k$, it is also true for $n=k+1$.

Step 3: If the statement is true for $n=1$ & so it is true for $n=2$ & so on. Hence it is true for all $n \geq 1$. (3 marks).



Question 26



SSHS

Prove by induction that $5^n + 3$ is divisible by 4 for any integer $n \geq 1$.

Prove $5^n + 3$ is divisible by 4

step 1 : Prove true for $n = 1$

$$5^1 + 3 = 8 \text{ which is divisible by 4}$$

$\therefore$ true for $n = 1$. ✓

Step 2 : Assume true for $n = k$.

$$\text{i.e. } 5^k + 3 = 4p \text{ for some integer } p$$

Step 3: Prove true for $n = k + 1$.

$$5^{k+1} + 3 = 5^{k+1} + 15 - 12$$

$$= 5(5^k + 3) - 12 \quad \checkmark$$

$$= 5 \times 4p - 12$$

$$= 4(5p - 3)$$

which is divisible by 4 ✓

$\therefore$ true for $n = k + 1$.

Hence if it is true for $n = k$, then it is true for $n = k + 1$. We have proved that it is true for $n = 1$, so it must be true for $n = 2$. If it is true for $n = 2$, then it must be true for $n = 3$, and so on. Hence it is true for all $n \geq 1$.



Question 27



ST GGHS

Use the process of Mathematical Induction to prove that for all positive integers n , where $n \geq 1$:

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

Test for $n = 1$

$$LHS = \frac{1}{3} \quad RHS = \frac{1}{2+1} = \frac{1}{3}$$

$\therefore LHS = RHS$ true for $n = 1$

Assume true for $n = k$

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1}$$

Prove true for $n = k + 1$

$$i.e. \frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2k+1)(2k+3)} = \left[\frac{k+1}{2k+3} \right]$$

$$LHS = \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)}$$

$$= \frac{k(2k+3)+1}{(2k+1)(2k+3)}$$

$$= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)}$$

$$= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$$

$$= \frac{(k+1)}{(2k+3)}$$

$\therefore$ true for $n = k + 1$

It has been proved true for $n = k + 1$, and from above it was proved true for $n = 1$

therefore it is true for $n = 1 + 1$, i.e. $n = 2$ and $n = 3$ and so on.

Therefore it is true for all integers $n \geq 1$



Question 28



ST GGHS

Prove by mathematical induction that $2^{3n} - 3^n$ is divisible by 5 for all positive integers n .

c) $2^{3n} - 3^n$ is divisible by 5

Prove true for $n=1$

$$2^3 - 3^1 = 8 - 3 = 5$$

$\therefore$ True for $n=1$

Let it be true for $n=k$

$$2^{3k} - 3^k = 5m \text{ where } m \text{ is a positive integer}$$

$$\therefore 2^{3k} = 5m + 3^k$$

Prove true for $n=k+1$

$$2^{3(k+1)} - 3^{k+1} = 2^{3k} \times 2^3 - 3^k \times 3^1$$

$$= (5m + 3^k) \times 8 - 3^k \times 3^1$$

$$= 40m + 8 \times 3^k - 3 \times 3^k$$

$$= 40m + 5 \times 3^k$$

$$= 5(8m + 3^k) \text{ which is}$$

divisible by 5 if m is a positive integer.

If it is true for $n=k$ we have proven it true for $n=k+1$. Since it is true for $n=1$, then it is true for $n=1+1=2$ and so on for all positive integral n .



Question 29



AMP

Prove by mathematical induction that:

$$4 + 4^2 + 4^3 + \dots + 4^n = \frac{4}{3}(4^n - 1)$$

(b) Step 1: Prove true for $n = 1$.

$$\text{LHS} = 4; \quad \text{RHS} = \frac{4}{3}(4 - 1) = 4 \checkmark$$

$\therefore \text{LHS} = \text{RHS}$

$\therefore$ True for $n = 1$.

Step 2: Assume true for $n = k$

$$\text{i.e. } 4 + 4^2 + 4^3 + \dots + 4^k = \frac{4}{3}(4^k - 1)$$

Required to prove that this is also true for $n = k + 1$.

$$\text{i.e. } 4 + 4^2 + 4^3 + \dots + 4^k + 4^{k+1} = \frac{4}{3}(4^{k+1} - 1)$$

Proof:

$$\begin{aligned} \text{LHS} &= 4 + 4^2 + 4^3 + \dots + 4^k + 4^{k+1} \\ &= \frac{4}{3}(4^k - 1) + 4^{k+1} \\ &= \frac{4}{3}(4^k - 1) + \frac{4}{3}\left(\frac{3}{4} \times 4^{k+1}\right) \checkmark \\ &= \frac{4}{3}(4^k - 1) + \frac{4}{3}\left(\frac{3 \times \cancel{4} \times 4^k}{\cancel{4}}\right) \checkmark \\ &= \frac{4}{3}(4^k - 1 + 3 \times 4^k) \\ &= \frac{4}{3}(4 \times 4^k - 1) \checkmark \\ &= \frac{4}{3}(4^{k+1} - 1) = \text{RHS as required.} \end{aligned}$$

Thus if it is true for $n = k$, then it is also true for $n = k + 1$.

Conclusion: It is true for $n = 1$, and it is true for $n = 2$, and so on for all positive integer values of n .



Question 30



BARKER COLLEGE

- (i) By considering the sum of an arithmetic series, show that

$$(1 + 2 + 3 + \dots + n)^2 = \frac{1}{4}n^2(n+1)^2 \quad 2$$

- (ii) Hence, use the Principle of Mathematic Induction to prove that

$$1^3 + 2^3 + 3^3 + \dots + n^3 = (1 + 2 + 3 + \dots + n)^2$$

for $n \geq 1$

4

6c) i) Consider $1 + 2 + 3 + \dots + n$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$\begin{aligned} S_n &= \frac{n}{2} [2 + (n-1)] \\ &= \frac{n}{2} (n+1) \end{aligned}$$

$$\begin{aligned} \therefore (1 + 2 + 3 + \dots + n)^2 &= \left[\frac{n}{2} (n+1) \right]^2 \\ &= \frac{1}{4} n^2 (n+1)^2 \end{aligned}$$

ii) let $n=1$

$$LHS = 1^3 = 1 \quad RHS = 1^2 = 1$$

$\therefore$ true for $n=1$.

Assume true for $n=k$.

$$\text{ie } 1^3 + 2^3 + 3^3 + \dots + k^3 = (1 + 2 + \dots + k)^2$$

Prove true for $n=k+1$

$$\text{ie } 1^3 + 2^3 + 3^3 + \dots + (k+1)^3 = (1 + 2 + \dots + (k+1))^2$$

$$\text{PROOF: } S_{k+1} = S_k + T_{k+1}$$

$$\begin{aligned} S_{k+1} &= (1 + 2 + \dots + k)^2 + (k+1)^3 \\ &= \frac{1}{4} k^2 (k+1)^2 + (k+1)^3 \\ &= \frac{1}{4} (k+1)^2 [k^2 + 4k + 4] \end{aligned}$$

$$= \frac{1}{4} (k+1)^2 (k+2)^2$$

$\therefore$ If true for $n=k$, it is true for $n=k+1$.

Since true for $n=1$, $\therefore$ true for $n=2, 3, \dots$

$\therefore$ By mathematical induction it is true for $n \geq 1$.



Question 31



CHS

Prove, by Mathematical Induction, that for $n \geq 1$

$$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \dots + \frac{1}{(3n-2) \times (3n+1)} = \frac{n}{3n+1}$$

c) When $n = 1$, LHS = $\frac{1}{4}$, RHS = $\frac{1}{3 \times 1 + 1} = \frac{1}{4}$

$\therefore$ true for $n = 1$.

Assume true for $n = k$, i.e.

$$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \dots + \frac{1}{(3k-2) \times (3k+1)} = \frac{k}{3k+1}$$

Now prove true for $n = k + 1$, i.e. prove that

$$S_k + T_{k+1} = S_{k+1}.$$

$$S_k = \frac{k}{3k+1}, S_{k+1} = \frac{k+1}{3k+4}, T_{k+1} = \frac{1}{(3k+1)(3k+4)}$$

$$\therefore S_k + T_{k+1} = \frac{k}{3k+1} + \frac{1}{(3k+1)(3k+4)}$$

$$= \frac{k(3k+4) + 1}{(3k+1)(3k+4)}$$

$$= \frac{3k^2 + 4k + 1}{(3k+1)(3k+4)}$$



Question 32



SAINT IGNATIUS

Prove by mathematical induction, where n is a positive integer and $n \geq 2$,

that $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$.

Prove $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$ for $n \geq 2$.

When $n=2$, LHS = $1 - \frac{1}{2^2} = \frac{3}{4}$

RHS = $\frac{2+1}{2 \times 2} = \frac{3}{4}$

$\therefore$ it is true for $n=2$.

Assume it is true for $n=k$:

$\therefore$ assume $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{k^2}\right) = \frac{k+1}{2k}$.

When $n=k+1$,

LHS = $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{k^2}\right)\left(1 - \frac{1}{(k+1)^2}\right)$

= $\frac{k+1}{2k} \left(1 - \frac{1}{(k+1)^2}\right)$ by assumption

= $\frac{k+1}{2k} \times \frac{(k+1)^2 - 1}{(k+1)^2}$

= $\frac{k^2 + 2k + 1 - 1}{2k(k+1)}$

= $\frac{k^2 + 2k}{2k(k+1)}$

= $\frac{k(k+2)}{2k(k+1)}$

= $\frac{k+2}{2(k+1)}$

= $\frac{n+1}{2n}$ when $n=k+1$.

$\therefore$ if it is true for $n=k$,
then it is true for $n=k+1$.

Since it is true for $n=2$,

it is true for $n=3, n=4, \dots$

i.e. it is true for all positive
integers $n \geq 2$



Question 33



ST GGHS

Prove by induction that

$$\sum_{k=1}^n k^2 = \frac{1}{6}n(n+1)(2n+1)$$

$$\sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1)$$

When $n=1$

$$\begin{aligned} \text{LHS} &= 1^2 & \text{RHS} &= \frac{1}{6}1(1+1)(2(1)+1) \\ & & &= \frac{1}{6} \cdot 1 \cdot 2 \cdot 3 \\ & & &= 1. \end{aligned}$$

∴ Statement is true for $n=1$.

Assume true for $n=k$.

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{1}{6}k(k+1)(2k+1)$$

Show for $n=k+1$

$$1^2 + 2^2 + 3^2 + \dots + (k+1)^2 = \frac{1}{6}(k+1)(k+2)(2k+3)$$

$$\begin{aligned} \text{LHS} &= 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 \\ &= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2 \\ &= \frac{1}{6}[k(k+1)(2k+1) + 6(k+1)^2] \\ &= \frac{k+1}{6}[k(2k+1) + 6(k+1)] \\ &= \frac{k+1}{6}[2k^2 + k + 6k + 6] \\ &= \frac{k+1}{6}[2k^2 + 7k + 6] \\ &= \frac{k+1}{6}[(k+2)(2k+3)] \\ &= \frac{1}{6}(k+1)(k+2)(2k+3) \\ &= \frac{1}{6}(k+1)[(k+1)+1][2(k+1)+1]. \end{aligned}$$

∴ If true for $n=k$, it is also true for $n=k+1$.

But since it is true for $n=1$, by induction it is true for all $n \geq 1$
 $n \in \mathbb{N}$.



Question 34



KINCOPPAL

Use the principle of Mathematical Induction to show that $7^n + 13^n$ is divisible by 10 for n positive odd integers.

Let $7^n + 13^n = 10M$ where M is any integer.

For $n=1$ $7^1 + 13^1 = 20 = 10 \times 2$ ✓
which is divisible by 10.

Assume $7^k + 13^k = 10M$ is true for $n=k$
Prove true for $n=k+2$

$$13^k = 10M - 7^k$$

$$7^{k+2} + 13^{k+2} =$$

$$\begin{aligned} 7^2 7^k + 13^2 13^k &= 7^2 7^k + 13^2 (10M - 7^k) \checkmark \\ &= 49 \cdot 7^k + 1690M - 169 \cdot 7^k \\ &= 1690M - 120 \cdot 7^k \\ &= 10(169 - 12 \cdot 7^k) \checkmark \end{aligned}$$

which is a multiple of 10, therefore true for $n=k+2$.

Since it is true for $n=1$, it is true for $n=1+2$
And so on, so it is true for all positive odd integers.